A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 22

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** $3.6 \times 10^{11} / 1.2 \times 10^{9}$ **1** = 300 years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** It must be porous, with spaces between the particles, so that it can store a large volume of water **1** It must be permeable, so that water can flow through it to a well or spring fast enough to be abstracted **1** An impermeable layer beneath it stops the water draining away, and the rock must be exposed at the surface somewhere, or overlain by permeable material, so that rainfall can recharge it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **permeable** → how easily water passes through / lets water through • **impermeable** → does not let water through	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction of 4.8×10^9 m^3 per year is four times the recharge of 1.2×10^9 m^3 per year, so water is being removed faster than it is replaced **1** The stored water is being mined rather than used sustainably, so the saturated zone shrinks and its upper surface, the water table, falls each year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer	2
01.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Wells and boreholes must be deepened and the water pumped from further down, so the energy cost of abstraction rises and shallow wells fail first **1** Rivers, springs and wetlands fed by the aquifer dry up, destroying habitats and the species that depend on them **1** Near the coast, sea water is drawn into the aquifer (saltwater intrusion), making the water too saline for drinking or irrigation **1** The rock compacts as the water that supported it is removed, causing land subsidence that damages buildings and drainage, and permanently reduces the storage capacity **1** Irrigated agriculture that depends on the aquifer fails, with loss of food production and livelihoods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Fossil water fell as rain thousands of years ago under a wetter climate – under the present climate recharge is negligible, so any abstraction depletes a finite store and the resource is effectively non-renewable **1** Changing to crops with a lower water demand, or to drip / trickle irrigation applied at night, which delivers water to the roots and loses far less to evaporation **1** Metering and pricing water, or setting abstraction licences and quotas, so that users have a reason to waste less **1** Recycling treated waste water for irrigation or industry, and harvesting rainwater, so less is taken from the aquifer **1** Lining canals and repairing leaking pipes so that less of the water abstracted is lost before it is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer • **Recycling** → reprocessing a material to make something new **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-121; AQA spec 3.2.2.1-3.2.2.2.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 **1** Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** 24-hour day / night cycle: people asleep overnight – morning rise as heating, lighting, cooking, industry and offices start **up to 2 Evening peak:** lighting, cooking, heating as people return home – short-term 'TV pickup' at programme breaks / mealtimes **up to 2 Other causes:** seasonal (winter heating / lighting, summer air conditioning) – weather-related (cold snaps) – weekday / weekend differences in industrial use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly **1** Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
02.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Intermittent renewable resources:** wind, solar, tidal **1** Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) **1** Breakdowns / maintenance of power stations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Potable:** no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints **1 Spray irrigation:** low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food **1 Condenser cooling:** no gross solids - pipe blockage **1 Boiler water:** no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter **1** Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Clay particles carry electrostatic charges that make them repel each other so they stay suspended **1** Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Chlorine **1** Ozone **1** Ultraviolet light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
05.1	1 mark for the point below. **1** As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2400 / 40 = 60 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Reserves:** the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) **1 Resource:** all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits **1** The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Lowering the cut-off ore grade includes deposits of lower purity in the reserves **1** Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	**1 mark each** 1 mark for each of the 3 points below. **1** The total quantity (stock / resource base) is far greater than could ever be used **1** But most is at purities too low or in locations too inaccessible for current economic exploitation **1** The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable – energy needs rise rapidly as ore grade falls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Human = 120 + 60 + 30 = 210 – total = 325 **1** 210 / 325 x 100 = 64.6% (accept 65%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • A bare wrong answer with no working.	2
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure – ammonia then converted to nitrates for fertiliser **1** Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO2 and NOx **1** Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	3
07.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Eutrophication of water bodies by leached nitrates / sewage **1** Photochemical smogs and tropospheric ozone from NOx **1** Acid rain (nitric acid) **1** Nitrous oxide as a greenhouse gas **1** NOx toxicity / respiratory disease **1** Nitrate in drinking water (methaemoglobinaemia) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Eutrophication** → nutrient enrichment of water leading to algal growth • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3

Do not write outside the box

Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is lost as CO2 (composting) or CH4 (digestion) so the C:N ratio improves / more nitrogen-rich **1** Bulk and odour reduced, pathogens killed, easier to apply – digestion also yields biogas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Mann-Whitney U test **1** Chi-squared test **1** t-test **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chi-squared compares frequencies / counts in categories **1** Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **chi-squared** → a test comparing observed frequencies with expected frequencies • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is always some remaining uncertainty / a probability that the result was produced by chance **1** Statistics only give a level of confidence (eg 95%, 99%) that the result is significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bars drawn above and below each mean showing plus and minus one standard deviation **1** If the bars of two means overlap a lot the difference is probably not significant – if they do not overlap it probably is **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) **1 But not proportional:** B has a high GDP but much lower energy use than A or E – C uses relatively high energy for its GDP **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes **1** Manufacture and transport of the extra products bought uses energy (indirect use) – larger houses use more energy to build **1** Affluent countries can afford energy for non-essential uses – cheap local energy (eg petrol in the USA) encourages wasteful use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Higher energy prices / taxes encouraging conservation **1** Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness **1** Milder climate needing less heating or air conditioning **1 Service-based economy:** heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted **1** Smaller / more efficient vehicles, better public transport, better insulated buildings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Decline of heavy industries such as steel and aluminium smelting in the UK **1** The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country **OR** efficiency improvements **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	1 mark for the point below. **1 Water:** pumping, treatment / sterilisation, sewage treatment – Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Smoke:** solid particulates (PM10, PM5, PM1) from incomplete combustion – small particles stay suspended, penetrate lungs – toxic chemicals (acids, phenols, metals) carried on particles – high albedo cools climate – blocks light reducing photosynthesis – damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths) – synergism with SO2 • **Controls:** Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion – smoke smogs have largely been eliminated in UK cities • **Acid rain gases:** SO2 (coal combustion, smelting sulfide ores) oxidised by ozone to SO3 - sulfuric acid – NOx (high temperature combustion) - nitric acid – HCl from PVC incineration – reactive gases travel long distances (transboundary pollution) • **Effects:** pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata) – dissolves exoskeletons – leaches Ca and Mg then mobilises toxic Al and Pb – corrodes metals, dissolves limestone – severity depends on bedrock (limestone neutralises) • **Controls:** fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays – switching from coal to gas / renewables	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Evaluation:** SO2 emissions in Europe have fallen dramatically and lake pH is recovering – NOx harder to control because of growing vehicle numbers – catalytic converters need lead-free fuel and warm engines – FGD is costly and produces wastes / gypsum – controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO2 vs NOx **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **particulates** → fine solid particles / smoke and dust • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NOx is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO_2) that caused it.	
10.2	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Reduces growth rates of trees / crops **1** Damages leaf cuticles (allowing SO2 to cause more damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

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